CSE 216 Formal Methods
Fall 2026 Daniel Fremont

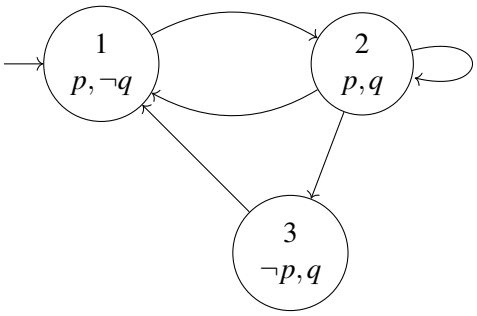
HW 6

Due November 10 at 3:00 pm

(2 questions, 55 points total)

1. (30 pts.) LTL and CTL

Consider the following Kripke structure M with atomic propositions p and q:



(a) Does $M \models GFq$? Why or why not?

Answer: yes. all states can always eventually end up in a state that has q.

(b) Does $M \models G(\neg q \, U \, p)$? Why or why not?

Answer: state 3 has q and $\neg p$, which means at this state the condition $\neg q \, U \, p$ is false

(c) Does $M \models G(q \rightarrow XXp)$? Why or why not?

Answer: no because from state 2 there exists a path where 2 steps from state 2 has $\neg p$. the path is 2 2 3.

(d) Does $M \models EGp$? Why or why not?

Answer: yes because there always exists a path from any node that has p hold globally, the one that loops forever on state 2.

(e) From which states does the formula EXq hold?

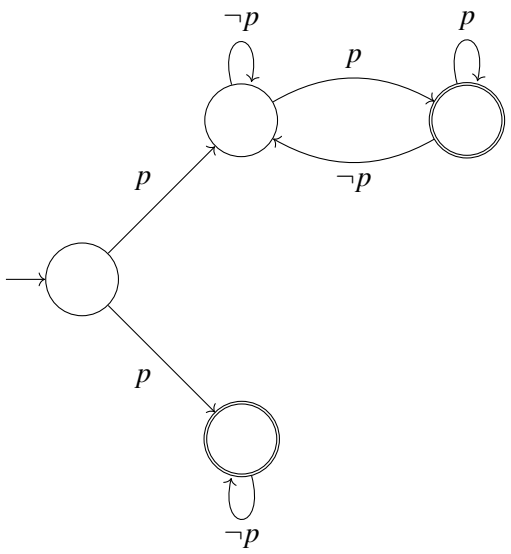
Answer: only 3 does not satisfy this since there is no immediate next state to which it can transition where q holds.

(f) Does $M \models AF(EXq)$? Why or why not?

Answer: yes. all states always finally have a state where their next state makes q true.

2. (25 pts.) Büchi Automata

Consider the following Büchi automaton B:



(a) Does B accept the trace $(p, \neg p, p, \neg p, \dots)$?

Answer: yes because the automata would visit the top accept state infinitely many times in this alternating trace.

(b) Does B accept the trace $(\neg p, \neg p, \neg p, \dots)$?

Answer: no because it would instantly reject. there isnt even an infinite trace starting with $\neg p$

(c) Does B accept the trace $(p, \neg p, \neg p, \neg p, \dots)$?

Answer: yes since the bottom branch would stay on the accepting node forever.

(d) Write an LTL formula satisfied by exactly the words accepted by B.

Answer: the bottom path is modeled by

$$p \wedge \mathrm{X}(\mathrm{G}\ \neg p)$$

and the top is modeled by

$$p \wedge \mathrm{G}\ \mathrm{F}\ p$$

so the total automata is

$$p \wedge (\mathrm{X}(\mathrm{G}\ \neg p) \vee \mathrm{G}\ \mathrm{F}\ p)$$